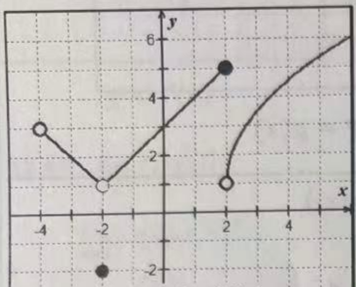
1.4 Limits of Composite Functions

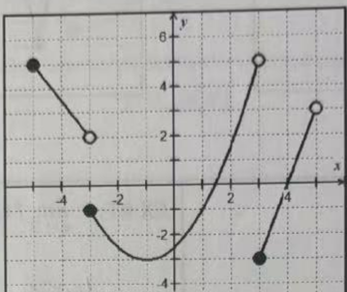
Name ____________________

Date ____________________ Period ________

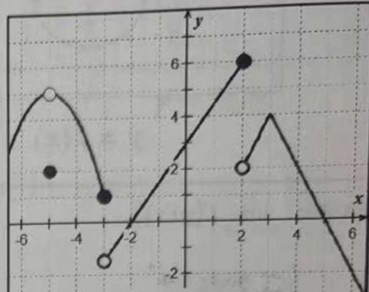
Problems 1 – 8, Use the graphs of $f(x)$, $g(x)$ and $h(x)$ shown below to answer the following limit questions.



Graph of f(x)



Graph of g(x)



Graph of h(x)

1. $\lim_{x \to -1} [f(x) + g(x)]$

$= 2 + (-3)$

$= -1$

2. $\lim_{x \to 0} [f(x) \cdot h(x)]$

$= 3 \cdot 3$

$= 9$

3. $\lim_{x \to -2} f(h(x))$

$= f(0)$

$= 3$

4. $\lim_{x \to 1} \sqrt{f(x)}$

$= \sqrt{4}$

$= 2$

5. $\lim_{x \to 0} h(|x| + 3)$

$= h(3)$

$= 4$

6. $\lim_{x \to -5^+} \left[\dfrac{g(x)}{h(x)}\right]$

$= \dfrac{5}{5}$

$= 1$

7. $\lim_{x \to -5} g(x + 2)$

$= g(-3)$

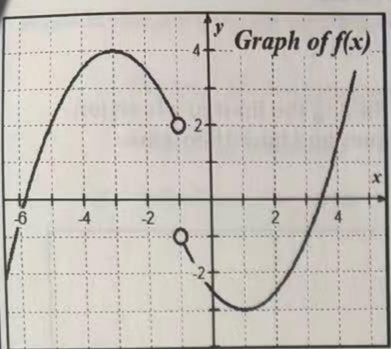
$=$ DNE

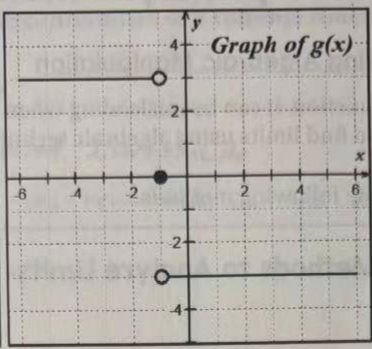
8. $\lim_{x \to -2} f\sqrt{x + 3}$

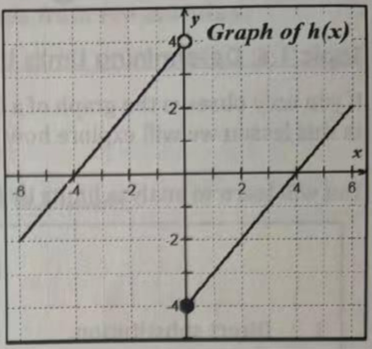
$= f(1)$

$= 4$

Problems 9 – 14, use the graphs of f, g and h below to evaluate the limits, if they exist.







9. $\lim_{x\to 5^+} g(25-x^2)$

$\lim_{x\to 5^+} g(25-x^2) = 0^-$

$\lim_{u\to 0^-} g(u) = -3$

$\lim_{x\to 5^+} g(25-x^2) = -3$

10. $\lim_{x\to -1} \left[(g(x))^2 - 5\right]$

$= \left(\lim_{x\to -1} g(x)\right)^2 - 5$

$= 9 - 5$

$= 4$

$\lim_{x\to -1^+} g(x) = -3$

$\lim_{x\to -1^-} g(x) = 3$

11. $\lim_{x\to 0} \frac{h(x)}{h(-x)}$

$\lim_{x\to 0^-} \frac{h(x)}{h(-x)} = \frac{4}{-4} = -1$

$\lim_{x\to 0^+} \frac{h(x)}{h(-x)} = \frac{-4}{4} = -1$

$\lim_{x\to 0} \frac{h(x)}{h(-x)} = -1$

12. $\lim_{x\to 0} h(h(x))$

$\lim_{x\to 0^-} h(x) = 4$

$\lim_{x\to 0^+} h(x) = -4$

$h(4) = 0$

$h(-4) = 0$

$\lim_{x\to 0} h(h(x)) = 0$

13. $\lim_{x\to -1} f(g(x))$

$\lim_{x\to -1^+} g(x) = -3$

$\lim_{x\to -1^-} g(x) = 3$

$f(-3) = 4$

$f(3) = -1$

$\lim_{x\to -1} f(g(x)) = DNE$

14. $\lim_{x\to -2} f\left(g(h(x))\right)$

$\lim_{x\to -2} h(x) = 2$

$g(2) = -3$

$f(-3) = 4$

$\lim_{x\to -2} f(g(h(x))) = 4$

Problems 15 – 16, Let $v(x) = \begin{cases} 3, & x = 0 \\ x, & x \neq 0 \end{cases}$ and $w(x) = \begin{cases} -3, & x < 0 \\ 0, & x = 0 \\ 3, & x > 0 \end{cases}$

15. $\lim_{x\to 0}(v(x)\cdot w(x))$

$\lim_{x\to 0^-}(v(x)\cdot w(x)) = 0\cdot -3 = 0$

$\lim_{x\to 0^+}(v(x)\cdot w(x)) = 0\cdot 3 = 0$

$\lim_{x\to 0}(v(x)\cdot w(x)) = 0$

16. $\lim_{x\to 0} \frac{w(x)}{v(x)}$

$\lim_{x\to 0^+} \frac{w(x)}{v(x)} = \frac{3}{0^+} = +\infty$

$\lim_{x\to 0^-} \frac{w(x)}{v(x)} = \frac{-3}{0^-} = +\infty$

$\lim_{x\to 0} \frac{w(x)}{v(x)} = +\infty$

1.5 Finding Limits by Analytic Methods

Name ______________________

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Problems 1 – 16, Find each of the following limits analytically. Show your algebraic analysis.

1. $\lim_{x\to 3}\left(\frac{2}{3}x^2+3x\right)$

$= \frac{2}{3}(3)^2+3(3)$

$= \frac{2}{3}\cdot 9+9$

$= 15$

2. $\lim_{t\to 4}\frac{t-4}{t^2-16}$

$= \lim_{t\to 4}\frac{\cancel{t-4}}{(t+4)\cancel{(t-4)}}$

$= \lim_{t\to 4}\frac{1}{t+4}$

$= \frac{1}{8}$

3. $\lim_{x\to -3}\frac{x^2-5x+6}{2x+6}$

$= \lim_{x\to -3}\frac{(x-2)(x-3)}{2(x+3)}$

$\lim_{x\to -3^+}\frac{(x-2)(x-3)}{2(x+3)} = \frac{(-)(-)}{(-)} = -\infty$

$\lim_{x\to -3^-}\frac{(x-2)(x-3)}{2(x+3)} = \frac{(-)(-)}{(+)} = +\infty$

$\therefore$ DNE

4. $\lim_{\theta\to\pi}[\sin^2\theta - 3\cos\theta]$

$(\sin\pi)^2 - 3\cos\pi$

$= 0-3(-1)$

$= 3$

5. $\lim_{x\to 2}\frac{x^3-8}{x-2}$

$= \lim_{x\to 2}\frac{\cancel{(x-2)}(x^2+2x+4)}{\cancel{x-2}}$

$= \lim_{x\to 2}(x^2+2x+4)$

$= 4+4+4$

$= 12$

6. $\lim_{\theta\to\frac{\pi}{3}}\frac{\tan\theta}{\theta^2}$

$= \frac{\tan\frac{\pi}{3}}{\left(\frac{\pi}{3}\right)^2}$

$= \frac{\sqrt{3}}{\frac{\pi^2}{9}}$

$= \frac{9\sqrt{3}}{\pi^2}$

7. $\lim_{x\to 0}\frac{\sqrt{x+5}-\sqrt{5}}{x}$

$= \lim_{x\to 0}\frac{x+5-5}{x(\sqrt{x+5}+\sqrt{5})}$

$= \lim_{x\to 0}\frac{1}{\sqrt{x+5}+\sqrt{5}}$

$= \frac{1}{2\sqrt{5}} = \frac{\sqrt{5}}{10}$

8. $\lim_{x\to 0}\frac{\frac{1}{3+x}-\frac{1}{3}}{x}$

$= \lim_{x\to 0}\frac{\frac{3-3-x}{(3)(3+x)}}{x}$

$= \lim_{x\to 0}\frac{-\cancel{x}}{(3)(3+x)\cancel{(x)}}$

$= \lim_{x\to 0} -\frac{1}{3(3+x)}$

$= -\frac{1}{9}$

9. $\lim_{x \to e} \frac{3x}{\ln x}$

$= \frac{3e}{\ln e}$

$= 3e$

10. $\lim_{x \to 2^+} \frac{3x^2+7x+2}{x^2-4}$

$= \lim_{x \to 2^+} \frac{(3x+1)\cancel{(x+2)}}{\cancel{(x+2)}(x-2)}$

$= \lim_{x \to 2^+} \frac{3x+1}{x-2}$

$= \frac{(+)}{(+)} = +\infty$

11. $\lim_{x \to 3} \frac{\frac{1}{x}-\frac{1}{3}}{x-3}$

$= \lim_{x \to 3} \frac{\frac{3-x}{3x}}{x-3}$

$= \lim_{x \to 3} -\frac{1}{3x}$

$= -\frac{1}{9}$

12. $\lim_{x \to \frac{3}{2}} \frac{8x^3-27}{2x-3}$

$= \lim_{x \to \frac{3}{2}} \frac{\cancel{(2x-3)}(4x^2+6x+9)}{\cancel{2x-3}}$

$= \lim_{x \to \frac{3}{2}} (4x^2+6x+9)$

$= 4 \cdot \frac{9}{4} + 6 \cdot \frac{3}{2} + 9$

$= 9+9+9 = 27$

13. $\lim_{x \to \pi^+} \cot x = +\infty$

14. $\lim_{x \to 0} \cos(x + \sin x)$

$= \cos(0+0)$

$= \cos(0)$

$= 1$

15. $f(x) = \begin{cases} 3x-1, & x \le 1 \\ 3x^2, & x > 1 \end{cases}$, find $\lim_{x \to 1} f(x)$

$\lim_{x \to 1^+} f(x) = 3x^2 = 3$

$\lim_{x \to 1^-} f(x) = 3x-1 = 2$

$\lim_{x \to 1^-} f(x) \neq \lim_{x \to 1^+} f(x)$

$\therefore \lim_{x \to 1} f(x) = DNE$

16. $\lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$

$= \lim_{h \to 0} \frac{\cancel{x^2}+2xh+h^2-\cancel{x^2}}{h}$

$= \lim_{h \to 0} \frac{\cancel{h}(2x+h)}{\cancel{h}}$

$= \lim_{h \to 0} (2x+h)$

$= 2x$